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A multi-objective optimization approach based on the Non-Dominated Sorting Genetic Algorithm II for power coordination in battery energy storage systems for DC distribution network applications

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ABSTRACT

Advancements in power electronics and renewable energy have transformed traditional distribution networks into active systems with distributed energy resources (DERs). DC networks provide an ideal platform for efficient DER integration, specifically Energy Storage Systems (ESS), underscoring the need for optimized coordination strategies. This study addresses the energy coordination problem in ESS within active DC distribution networks. In such networks, ESS must be efficiently managed due to daily demand variability and solar generation fluctuations, with proactive strategies essential to reduce operational costs and energy losses. The proposed methodology uses a multi-objective nonlinear programming approach, resolved through a master–slave framework. In the master stage, the non-dominated genetic algorithm (NSGA-II) is employed to determine ESS charge and discharge actions, while in the slave stage, an hourly power flow method based on successive approximations assesses the objective functions. The methodology was validated on a 33-node DC test system adapted to the generation and demand conditions of Medellín, Colombia. MATLAB simulations demonstrate that NSGA-II outperforms other methods in Pareto front quality, best solution, average response, and standard deviation. Results show that NSGA-II is the optimal choice for minimizing operational costs and losses in DC networks, confirming its applicability and effectiveness within the studied context. This methodology represents an advancement in ESS management for distribution networks, particularly in applications where anticipating demand and generation variability is critical.

1. Introduction

1.1. General context

In recent years, advances in power electronics, renewable energy generation, energy storage systems, and electrical transmission and distribution technologies have revolutionized electrical distribution networks, transitioning them from passive structures to complex active topologies [1]. Unlike passive networks, which traditionally depend on a single active source supplying power to downstream consumers, active distribution networks incorporate multiple distributed energy resources (DERs) across the grid. This paradigm shift has introduced bidirectional power flows and fundamentally altered economic interactions between grid users and utility companies [2,3].

The integration of DERs — such as renewable energy sources and energy storage systems — primarily occurs through power electronic converters, as illustrated in Fig. 1. These conversion systems enable the interconnection of diverse energy resources with both AC and DC distribution networks, allowing independent control of active and reactive power in AC systems or the exclusive management of active power

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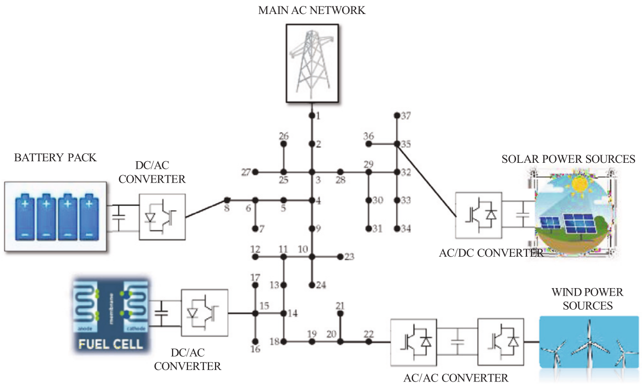

Fig. 1. Schematic connection of distributed energy resources in AC distribution networks.

in DC networks [4–6]. Such flexibility in power control is essential for maintaining voltage stability and minimizing power losses, particularly as DER penetration rates increase.

Increasingly, DC-operated distribution networks are gaining attention in industrial applications and academic research due to their efficiency advantages over traditional AC networks. DC networks exhibit reduced energy losses and improved voltage profiles [7]. Additionally, DC systems simplify control by focusing solely on voltage regulation, unlike AC networks that require coordinated control of both voltage and frequency [8]. Furthermore, since most DERs (e.g., solar photovoltaic systems and energy storage devices) generate or store power in DC form, using DC distribution networks reduces the need for conversion stages. This not only enhances network reliability but also improves overall system efficiency by eliminating complex power inverter stages and their associated switching losses [9,10].

1.2. Motivation

The integration of distributed energy resources (DERs) in modern electrical networks presents both opportunities and challenges. While DERs—such as energy storage systems (ESS)—offer enhanced reliability, flexibility, and sustainability, their effective coordination remains a significant challenge, particularly in DC distribution networks. DC networks, as opposed to their AC counterparts, allow for reduced energy losses, improved voltage profiles, and simpler control requirements. These advantages, combined with the direct compatibility of DC technologies with most DERs, underscore the potential for DC networks to drive more efficient and resilient power systems [11].

In this context, it is crucial to develop strategies that optimize the performance of ESS within DC networks. Effective power coordination of ESS can lead to substantial reductions in operational costs at the grid level by minimizing power generation requirements at substations. Furthermore, optimized ESS operation can help reduce overall power losses across distribution branches, thereby enhancing the efficiency and reliability of the entire network. This research is motivated by the need to address these critical issues through a comprehensive energy management system tailored for DC networks with multiple DERs, as depicted in Fig. 2. By focusing on cost and loss minimization, this study seeks to provide a robust foundation for advancing ESS coordination in next-generation DC distribution networks

1.3. Literature review

In the specialized literature, multiple optimization approaches have been proposed for the optimal dispatching of ESS in electrical networks to minimize single or multi-objectives. Some of these approaches are presented below. Authors of [12] present the usage of ESS in low-voltage distribution networks to compensate for over-voltages caused by photovoltaic sources, which typically limits all the potentiality of integrating renewable energy resources in low-voltage networks. In this research, the ESS are used to store energy when solar sources inject high levels of power, avoiding its direct injection into the distribution networks, preventing over-voltage, and returning the energy storage in ESS when solar energy is unavailable and demand increases. Authors of [13] propose a multi-objective optimization approach for minimizing greenhouse emissions and reducing voltage deviations via the optimal planning and operation of renewable energy and storage technologies using a multi-period analysis environment. The proposed solution methodology corresponds to the multi-objective multi-verse optimizer, and the authors validate their proposal in a 68-node test feeder. In Ref. [14], a convex approximation via a recursive programming approach was proposed to minimize the expected power purchasing costs in electrical networks, including diesel sources, by coordinating ESS and renewables in a day-ahead operation scenario. Numerical validations in the 33-bus grid confirmed the effectiveness of the solution approach compared to the exact nonlinear programming model and with metaheuristics, such as the Salp swarm optimizer and the Crow search algorithm, among others.

Authors of [15] presented a multi-objective analysis for coordinating power in ESS in virtual power plant applications. The main idea of this proposal was to minimize the energy production costs and the network impact caused by the load spikes. The authors' main contribution is integrating the multi-objective tool optimizer in the DIgSILENT power system simulation tool. This software effectively models the electrical variables' performance in large-scale realistic low-voltage networks, considering a multi-time domain analysis using a daily operative scenario. The authors used the non-sorted second-generation genetic algorithm (NSGAII) as the multi-objective optimizer approach. In Ref. [16], the authors proposed the ϵ-constraint method to formulate a multi-objective optimization problem for the optimal power coordinating in ESS and renewable generators considering a demand response operative environment. The general algebraic modeling system software and its nonlinear optimization tools were used in the numerical simulation using the 33-bus grid to minimize the expected installation and operation costs in distributed energy resources while improving a voltage stability index. Authors of [17] propose a power coordination strategy between ESS and electric vehicles using a multi-objective random black-hole particle swarm optimization algorithm. The authors proposed a multi-objective optimal peak load-shaving strategy to achieve the best peak load-shaving effect with the minimum electricity cost. Comparative analysis with the classical multi-objective particle swarm optimizers showed additional improvements in the final Pareto front in favor of the proposed approach.

Authors of [18] proposed a rule-based peak-shaving method using master–slave optimization to reduce fuel consumption in an electrical microgrid with an isolated diesel generator. An ESS manages generator power by scheduling charge/discharge based on day-ahead power demands. The slave optimization level calculates optimal low-voltage AC bus power, while the master optimization level determines optimal distributed generator control inputs. This research's main idea is to manage power in the diesel generator and the ESS optimally to supply the power requirements of a low-voltage distribution microgrid while maintaining a secure diesel generator operation between its operative ranks. In Ref. [19], a multi-objective particle-swarm optimizer (PSO) was proposed to manage power in electrical DC distribution networks with high solar power resources and ESS penetration. The PV sources were operated applying the maximum power point tracking concept, and the proposed multi-objective PSO approach simultaneously minimizes energy losses and grid operating costs, i.e., energy purchasing and maintenance costs in ESSs and PVs. Numerical results in the 33-bus networks demonstrated the effectiveness of the proposed multi-objective (PSO) approach compared to a multi-objective version of the ant lion optimizer (ALO).

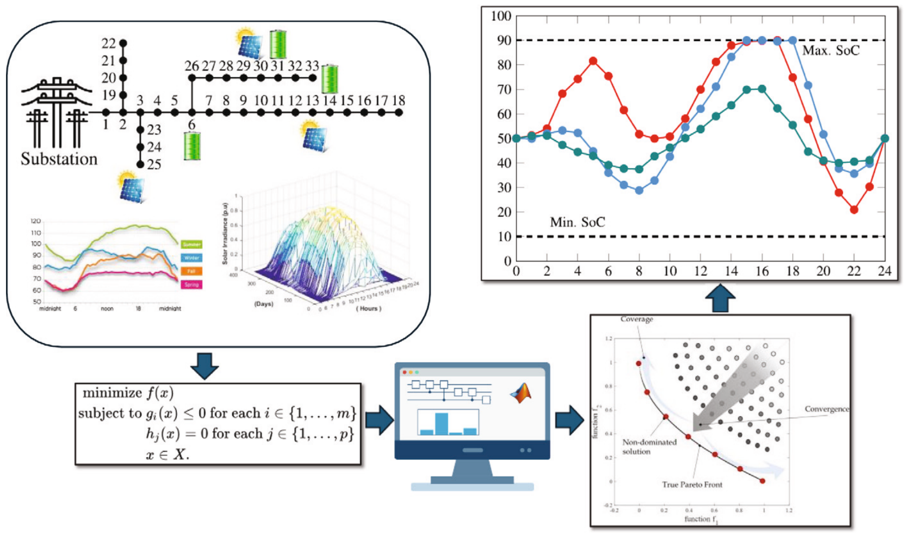

Fig. 2. Energy management system in electrical networks considering multi-objective analysis.

In [20], energy management in rural and urban DC microgrids with ESS is addressed by optimizing operational costs, energy losses, and CO_2 emissions through a multi-objective model. This document employs various multi-objective optimization algorithms and a power flow method based on successive approximations, allowing for a precise analysis of operational constraints. Validated in 27- and 33-node scenarios in Colombia, with photovoltaic generators and batteries, it demonstrates its applicability in local conditions. However, this document neither presents nor analyzes the Pareto front, making it more difficult to assess the relative performance of the methods used in the study. The absence of this analysis limits the flexibility to adapt solutions to different scenarios or needs. The study presented by [21] addresses the increasing complexity in managing microgrids that integrate DERs and electric vehicle (EV) charging stations in response to the demand for clean and affordable energy. A multi-objective optimization model is proposed to reduce operating costs and emissions, taking into account the load of ESS and EV charging stations within microgrids that include technologies such as photovoltaic panels, fuel cells, and wind turbines. The model uses day-ahead energy scheduling and three optimization scenarios: minimizing costs and maximizing BSS benefits, minimizing emissions, and a combination of both objectives. Using the Krill Herd algorithm, the strategy achieves a balance between profitability and sustainability and demonstrates, compared to other methods, that integrating EV charging infrastructure and renewable energy into microgrids improves operational efficiency and reduces environmental impact. However, by having specific analysis scenarios (weighted objective function), decision-making is limited, as solutions that better balance conflicting objectives are unknown. Additionally, the mathematical model assumes a single-node network rather than a multi-node network, as is typically the case with distribution systems.

The authors of [22] address the optimal design and operation of microgrids by integrating technical, economic, and environmental aspects into a single decision-making framework. Through a multi-objective mixed-integer linear programming algorithm, they minimize the levelized cost of energy and life cycle emissions for microgrids with photovoltaic systems and ESS, either connected or disconnected from the grid. The model considers a 25-year planning horizon and hourly energy management, ensuring long-term planning combined with real-time operation. In a case study based on the French electricity grid, a Pareto front is generated to illustrate the trade-off between costs and emissions, and to assess the impact of limiting grid subscription power on system efficiency and local renewable energy utilization. However, as in the previous case, the authors simplify the network to a single-node system, disregarding network power losses. In [23], a multi-objective dispatch is proposed for energy storage systems (ESS) and renewable energy resources (RES) in distribution networks, using a convex weighting-based approach. To ensure a global optimum, the nonlinear model is converted into a second-order equivalent, leveraging convex properties in the complex-variable domain. The model enables the construction of the optimal Pareto front, addressing objectives such as minimizing grid operating costs and energy losses. Numerical results on IEEE 33- and 85-bus networks confirm the effectiveness of the proposed model compared to algorithms like the continuous genetic algorithm, particle swarm optimizer, and vortex search algorithm, demonstrating that the proposed approach is suitable for AC distribution networks.

On the other hand, most of the analysis focuses on single-objective function analysis regarding renewable energy resources and ESS in DC networks. These objectives can be technical, economic, or environmental indexes. In the case of technical objective functions, energy power losses [24], stability indexes [25], or voltage profile improvement [26] are the most common objectives. The most used economic objective functions are energy purchasing costs minimization [27] or minimization of the energy losses costs [28]. In contrast, in the case of the environmental objectives, most research focuses on minimizing the expected carbon dioxide emissions [29,30].

1.4. Novelty in relation to other studies

This work introduces distinct advancements compared to existing studies in the field:

1. **Focus on DC Networks:** While most studies focus on AC networks, this research specifically targets the coordinated management of ESS in DC distribution networks. Unlike prior works that address single objectives (e.g., costs or losses) in DC networks, this study simultaneously optimizes operational costs and energy losses, leveraging the inherent advantages of DC networks, such as simpler control requirements and reduced conversion losses.
2. **Integrated Methodology with NSGA-II:** This work implements a *master–slave* framework where, in the *master* stage, NSGA-II generates ESS configurations, and in the *slave* stage, a power flow method validates these configurations under operational constraints. This integration achieves superior Pareto front quality and balance among conflicting objectives, surpassing studies that use alternative metaheuristic algorithms like MOALO or MOPSO.

3. **Validation with Real Data and Comprehensive Network Configuration:** Unlike other studies that simplify networks to single-node models or idealized scenarios, this research employs a 33-node test system with real demand and solar generation data from Medellín, Colombia. Additionally, the study includes comprehensive ESS characteristics, such as charge/discharge limits and state of charge, providing a realistic and practical representation for industrial applications.
4. **Comparison with Established Metaheuristics:** This study conducts a detailed comparison of NSGA-II against metaheuristics like MOPSO and MOALO. Results show that NSGA-II outperforms these alternatives in terms of solution diversity, lower standard deviation, and better balance between costs and losses. This level of quantitative analysis is uncommon in the literature and highlights the originality of this approach.

This work not only expands the scope of ESS management in DC networks but also establishes a robust methodological benchmark validated with real-world data, paving the way for future research in the field.

1.5. Contributions of the document

This work presents significant advancements in both academic and industrial domains, highlighting the following key contributions:

1. **Multi-objective Optimization Framework for DC Distribution Networks:** A novel methodology based on a *master–slave* approach is introduced to simultaneously minimize operational costs and energy losses in DC distribution networks. In the *master* stage, the NSGA-II algorithm is employed to generate optimal charging and discharging configurations for ESS. In the *slave* stage, an hourly power flow method using successive approximations evaluates operational constraints and objective functions. By incorporating daily demand and solar generation variability, this approach ensures robust and practical solutions for active network management.
2. **Impact on Scientific Literature:** This work extends the current knowledge on ESS management in DC networks, a domain significantly less explored compared to AC networks. The proposed methodology simultaneously optimizes technical and economic objectives, addressing gaps in the literature where these aspects are often treated in isolation. Moreover, the NSGA-II algorithm demonstrates superior performance in Pareto front quality, solution diversity, and balancing conflicting objectives when compared to metaheuristic techniques such as MOALO and MOPSO.
3. **Relevance to the Electric Industry:** The proposed methodology equips electrical system operators with an adaptable tool to enhance operational efficiency and reduce costs in distribution networks with high penetration of distributed energy resources. By combining technical and economic optimization, this framework enables intelligent ESS management, maximizing their utility during periods of solar variability and energy demand, thus ensuring more economical, efficient, and sustainable operations.

These contributions establish this research as a significant milestone in advancing management strategies for DC distribution networks, offering practical and robust solutions for both academic and industrial applications.

In terms of scope, this research is based on the following premises: (i) the demand profile and solar generation data are considered constant inputs derived from historical data and statistical analysis; (ii) the locations and capacities of the solar sources and ESS are predetermined by the distribution system operator during a planning phase, thus treating these variables as fixed within this operational optimization context; and (iii) the choice of NSGAII as the optimization methodology is supported by its demonstrated effectiveness in handling multi-objective problems across scientific and engineering disciplines [31–33].

This work builds upon and enhances numerical results reported in [19], where the authors previously applied multi-objective PSO and ALO methods to the same 33-bus network for joint minimization of network energy losses and operating costs. Improvements achieved in this study can be observed in the distribution of Pareto-optimal solutions and in achieving lower standard deviations in the objective metrics.

The choice of NSGAII for solving this multi-objective problem is further justified by its robustness, attributed to its non-dominated sorting, elitism, and crowding distance mechanisms, which collectively make it one of the most effective tools for multi-objective optimization [34]. Moreover, NSGAII's ability to produce a diverse set of Pareto-optimal solutions in a single run provides a significant edge over traditional methods [35], positioning it as a preferred algorithm for tackling complex multi-objective challenges across diverse domains [36].

1.6. Document structure

The remainder of the components of this article are structured as follows: Section 2 describes the general optimization problem associated with the multi-objective power coordination in ESS for DC distribution network applications. The objective functions considered correspond to the simultaneous minimization of energy losses and energy purchasing costs in a daily operation scenario. Section 3 describes the solution methodology which is based on the NSGAII approach. Section 4 describes the main characteristics of the 33-bus network in its DC version, including the information regarding daily demand and solar behaviors as well as the ESS' characteristics. Section 5 shows the numerical analyses and discussions using a comparative methodology with the MOPSO and the MOALO. Finally, Section 6 list the main concluding remarks derived from this research as well as some possible future works.

2. Mathematical formulation

This section presents the mathematical formulation used to represent the power coordination problem of ESS in DC grids, considering an environment of daily demand and solar generation variations under a day-ahead operation scenario. This mathematical formulation was adapted from [30].

2.1. Objective functions

In this work, the objective is to simultaneously minimize the cost of energy purchase and the energy losses of the power system for a single day of operation. It should be noted that these two objective functions are in conflict with one another in the operation of the network. Consequently, it is necessary to employ multi-objective strategies in order to resolve this problem [37].

$$OF_1 = min\left(\sum_{t=1}^{H} C(t)\right) \quad (1)$$

$$C(t) = C_{EE}(t) + C_M(t) \quad (2)$$

The first objective function is related to the minimization of the energy operational costs of the DC grid for an operation day ($Ecosts$), which are calculated by using Eq. (1), where H denotes the set of periods analyzed (in this particular case 24 h). In each period, the operational cost of the DC grid ($C(t)$) is calculated by considering the energy purchasing costs (C_{EE}) and the maintenance costs of the distributed energy resources(C_M).

$$C_{EE}(t) = \sum_{i\in N}\left(C_{CGi}(t) * p_{CGi}(t)\right) \quad (3)$$

$$C_M(t) = \sum_{i\in N}\left(C_{MDGi} * p_{DGi}(t) + C_{MBi} * p_{Bi}(t)\right) \quad (4)$$

Eq. (3) allows the calculation of C_{EE}, this variable is composed by the energy production cost (C_{CGi}) and the power generated (p_{CGi}) by the conventional generators installed in i_{th} bus of the network, where N represents the set of buses that compose the DC grid. Eq. (4) calculates the maintenance cost of the distributed generators and ESS installed in the electric grid. In this equation, C_{MDGi} and C_{MBi} are the maintenance costs by kwh of the distributed generators (DGs) and ESS. While $p_{DGi}(t)$ and $p_{Bi}(t)$ are the power supply and storage by the DGs and ESS, respectively.

$$OF_2 = min\left(\sum_{t=1}^{H} p_{loss}(t)\right) \tag{5}$$

The second objective function employed in this paper is the minimization of energy losses in the DC grid associated with the transport of energy. Eq. (5) summarizes all the energy losses obtained in a day of operation.

$$\begin{aligned} p_{loss}(t) &= \sum_{i\in N}\left(p_{CGi}(t) + p_{DGi}(t) \pm p_{Bi}(t) - p_{di}(t)\right) \\ &= \sum_{i\in N}\left(\sum_{j\in N} G_{ij} * v_i(t) * v_j(t) - G_{i0} * V_i^2(t)\right) \end{aligned} \tag{6}$$

For calculating the energy losses in each period, it is used Eq. (6). In this Equation $p_{di}(t)$ denotes the power demanded in the bus i, G_{ij} is the conductance value of the line that interconnected the bus i and j, G_{io} is the conductance of the impedance loads connected to the bus i, $v_i(t)$ and $v_j(t)$ are the voltage profiles of the bus i and j, respectively.

2.2. Set of constraints

In this subsection, all constraints that represent the operation of a DC grid are formulated from Eqs. (7) to (17) .

$$\begin{aligned} &P_{CGi}(t) + P_{DGi}(t) \pm P_{Bi}(t) - P_{di}(t) \\ &= V_i(t) * \sum_{j\in N} G_{ij} * V_j(t) \qquad \forall i,j \in N,\ \ \forall t \in H \end{aligned} \tag{7}$$

$$P_{CGi}^{min} \leq p_{CGi}(t) \leq P_{CGi}^{max} \qquad \forall i \in N,\ \forall t \in H \tag{8}$$

$$P_{DGi}^{min} \leq p_{DGi}(t) \leq P_{DGi}^{max} \qquad \forall i \in N,\ \forall t \in H \tag{9}$$

Eq. (7) represents the power balance in the DC network, while the equations from (7) to (10) guarantee the power limits of conventional generators and DGs. Where min and max denote the minimum and maximum power limits, respectively.

$$P_{Bi}^{char\_max} \leq p_{Bi}(t) \leq P_{Bi}^{disch\_max} \qquad \forall i \in N,\ \forall t \in H \tag{10}$$

$$P_{Bi}^{char\_max} = -\frac{C_{Bi}}{tc_{Bi}} \qquad \forall i \in N \tag{11}$$

$$P_{Bi}^{disch\_max} = \frac{C_{Bi}}{td_{Bi}} \qquad \forall i \in N \tag{12}$$

In the particular case of ESS, the power limits are defined by the maximum charge ($P_{Bi}^{char\_max}$) and discharge ($P_{Bi}^{disch\_max}$) power, which is defined by the type of ESS installed on the bus i (i.e., datasshet). To calculate these limits, the capacity of the ESS and its charge and discharge time are used (see Eqs. (11) and (12)).

$$SOC_i^{\min} \leq SOC_{Bi}(t) \leq SOC_i^{\max} \qquad \forall i \in N,\ \forall t \in H \tag{13}$$

The ESS must control the state of charge (SOC) and satisfy Eq. (13) at all times. In this equation, $SOC_i^{\min}$ and $SOC_i^{\max}$ refer to the minimum and maximum state of charge of the ESS. To update the SOC in each time period Eq. (14) is used, in this formulation ϕ_{Bi} is the charge factor of the BESS, which is a function of the power limits and the charge and discharge times (see Eq. (15)). While, $\Delta(t)$ is associate to the duration of the charging or discharging action of the ESS in the period of time t, in this research $\Delta(t) = 1$.

$$\begin{aligned} &SOC_{Bi}(t) = SOC_{Bi}(t-1) - \left(\phi_{Bi} * P_{Bi}(t)\right) * \Delta t \\ &\forall i \in N,\ \forall t \in H \end{aligned} \tag{14}$$

$$\phi_{Bi} = \frac{1}{tc_{Bi} * P_{Bi}^{char\_min}} = \frac{1}{td_{Bi} * P_{Bi}^{disch\_max}} \qquad \forall i \in N,\ \forall t \in H \tag{15}$$

It is important to highlight that the initial and final SoC could be controlled for obtaining the best performance of the ESS. In this way, this paper has established an initial and final SOC of 50% following the suggestion made in [38]. Note that the ESS model adopted in this document (Eqs. (10)–(15)) is a simplified model that relates the SoC of the ESS to the charging and discharging power through a linear coefficient (ϕ_{Bi}), as shown in [39]. This simplification avoids the use of binary variables to distinguish between charging and discharging, as presented in [40]. However, this simplified model does not fully capture the rigorous operational dynamics of ESS, but it allows to work with a nonlinear programming model instead of a mixed integer nonlinear programming model.

$$V_i^{\min} \leq v_i(t) \leq V_i^{\max} \qquad \forall i \in N,\ \forall t \in H \tag{16}$$

$$i_{ij}(t) = \frac{\left|v_i(t) - v_j(t)\right|}{R_{ij}} \leq I_{ij}^{\max} \qquad \forall i,j \in N,\ \forall t \in H \tag{17}$$

In relation to the constraints associated with the operating conditions of the DC grid, in this research are included the Eqs. (16) and (17). The limits of the voltage profiles are set in Eq. (16) where V_i^{min} and V_i^{max} are the minimum and maximum voltage limits. The current limits are set by Eq. (17), where $i_{ij}(t)$ is the line current at time t of the line connecting bus i and j, and R_{ij} is the line resistance.

3. Solution methodology

This section describes the codification and proposed solution for solving the problem of power coordination of ESS in DC grids to reduce operating costs and energy losses. This section starts by presenting the used codification. Followed by the fitness function and the proposed master–slave strategy between the NSGAII [41] and the hourly power flow based on successive approximations (HPFSA) [42].

3.1. Codification

To address the problem of power coordination of ESS in DC grids a continuous vector encoding of size $[1, N_B \cdot |H|]$ is proposed. Where, N_B refers to the number of ESS available in the DC distribution network, while $|H|$ corresponds to the number of periods considered under the horizon time analyzed. Each position of the encoding vector takes a value in the range between 0.1 and 0.9, representing the SOC of ESS for each period. The SOC is taken as a decision variable in the optimization problem since it directly influences the power generated or absorbed by the ESS (Eq. (14)). In the proposed solution strategy, the SOC determines the capacity of the ESS to generate or absorb power at any given time, directly affecting the active power balance at the system nodes (Eq. (7)). This means that the SOC is not a fixed or assumed value but varies dynamically according to the system's operational conditions. This allows the proposed methodology to intelligently manage the power generated by distributed generators.

It is noteworthy that, for both the initial and final time instants of each analyzed period, the encoding takes the value of 0.5, following the constraints imposed for the ESS. This ensures that for each ESS, during the analysis period, the charge starts and ends at 50% of its nominal capacity. Fig. 3 depicts the aforementioned encoding.

3.2. Fitness function

With the aim of improving the exploration and exploration of the proposed NSGAII. In this paper it has been used a fitness function (FF) that allows to explore infeasible solutions with the aim of reducing the processing times and improving the quality of the solution obtained by the NSGAII. This FF employs a penalty factor (PF) to impose a penalty

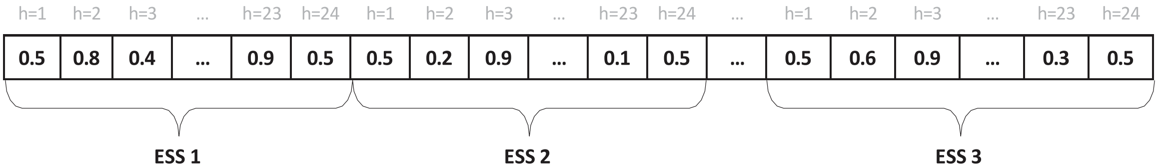

Fig. 3. Proposed coding to resolve the power coordination problem of ESS in DC grids.

on the objective functions utilized when a constraint is violated, in other case the penalty factor takes a value of zero. The penalty factor is controlled by the constant $\beta = 1000$ for guarantee that the units of the PF impact the objective functions in the case of a violations. The Eqs. (18) and (19) represent the FF and PF used.

$$FF = OF_i + \beta * PF \qquad \forall i \in 1, 2. \tag{18}$$

$$PF = \sum_{t=1}^{24} \left(\begin{array}{l} max\left\{0, \sum\limits_{i\in\mathcal{N}} (p_{Bi}(t) - P_{Bi}^{char\_max})\right\} \\ + \left| min\left\{0, \sum\limits_{i\in\mathcal{N}} (p_{Bi}(t) - P_{Bi}^{disch\_max})\right\}\right| \\ max\left\{0, \sum\limits_{i\in\mathcal{N}} (SOC_{Bi}(t) - SOC_i^{\max})\right\} \\ + \left| min\left\{0, \sum\limits_{i\in\mathcal{N}} (SOC_{Bi}(t) - SOC_i^{\min})\right\}\right| \\ + max\left\{0, \sum\limits_{i\in\mathcal{N}} (v_i(t) - V_i^{\max})\right\} \\ + \left| min\left\{0, \sum\limits_{i\in\mathcal{N}} (v_i(t) - V_i^{\min})\right\}\right| \\ + max\left\{0, \sum\limits_{i\in\mathcal{N}} \sum\limits_{j\in\mathcal{N}} (i_{ij}(t) - I_{ij}^{\max})\right\} \end{array} \right) \tag{19}$$

3.3. Master-slave strategy

As solution strategy, this article proposes a two-stage methodology (i.e., master–slave), where the first stage uses a multi-objective optimization method known as NSGA II. This generates charge and discharge configurations (i.e., power coordination) based on the SOC of each ESS in the different periods of time analyzed for improving two objective function in conflict (i.e., energy operational costs and energy losses). In the second stage, an power flow method is implemented for each SOC configuration proposed for the master-stage with the aim to evaluate the fitness functions using the HPFSA. The HPFSA allows to evaluate the fitness function of each configuration based on its result, and to determine whether the evaluated solution is feasible or infeasible, penalizing infeasible solutions. Through an iterative process, high quality solutions to the problem of power coordination of ESS in DC grids are obtained.

3.3.1. Master stage: Non-dominated sorting genetic algorithm II

For the first stage of the methodology, the multi-objective optimization method denominated NSGA II is implemented. This algorithm, proposed in [41], is a modification of the Population-based Genetic Algorithm (PGA) proposed in [43]. NSGA II combines the genetic operators of PGA with a sorting stage of non-dominated solutions into Pareto fronts, allowing the simultaneous evaluation of more than one objective function. This is based on the assumption that these objective functions are in conflict, i.e., improving one means worsening the others and vice versa. This allows a set of high-quality solutions to be determined at each iteration, in contrast to the single-objective approach, which produces only one solution in each generation cycle. In this paper, for the sorting process of non-dominated solutions, the objective functions described in Eqs. (1) and (5) are considered simultaneously.

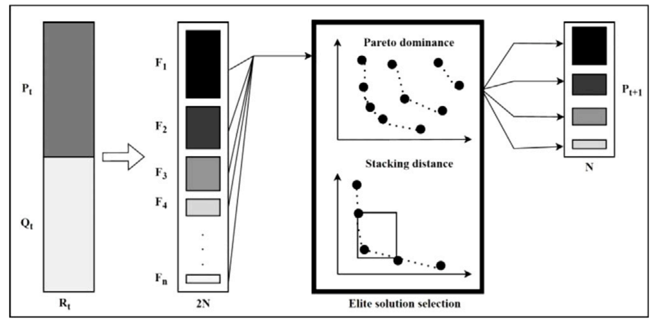

Fig. 4. Non-dominated sorting genetic algorithm II.

To initialize the NSGAII algorithm, we proceed with the generation of an initial population P of N individuals, which can be generated randomly or by means of a heuristic initialization technique. From P and making use of the genetic operators of selection, crossover and mutation, we proceed to create a second population of individuals called Q of the same size as P. To implement the selection genetic operator, it is possible to use any of the objective functions of the problem under analysis as metric, for this case the objective function described in Eq. (1) was used and selection by tournament was performed. The next step of the NSGA II algorithm is to combine the two populations P and Q into a single population R, which will have a size of $2 \cdot N$ individuals. The population R must be sorted into Pareto fronts (i.e. F_1, F_2, $F_3 \cdots F_n$), for which, the concept of Pareto dominance is used, which specifies that an individual X_1 dominates another individual X_2 (both belonging to the population R) if the following two rules are satisfied:

- Individual X_1 is not worse than individual X_2 in all considered objective functions.
- Individual X_1 is strictly better than individual X_2 in at least one of the addressed objective functions.

Once the classification of all individuals of the population R into Pareto fronts is done, we continue to select individuals from the best fronts until a number of N individuals are complete. For this purpose, the best Pareto front is considered to be F_1. These best individuals are those that make up the population P^{t+1}. Note that not all constructed fronts can be part of the new population P^{t+1}, so these fronts must be eliminated. Likewise, there is the possibility that when considering the last front (i.e., F_i) to complete the N individuals of P^{t+1}, not all individuals can be included because they exceed the size of N. In this case, a strategy must be considered to select some individuals from F_i (until the completion of N individuals). For this case we used the method of stacking distance, which can be consulted in [44]. From this point on, the method is repeated until the maximum number of iterations t_{max} is reached. Finally, F_1 contains the set of best quality solutions found by the algorithm. Fig. 4 present a overview of NSGA II algorithm and Fig. 5 illustrates the steps of the NSGA II algorithm proposed in this paper.

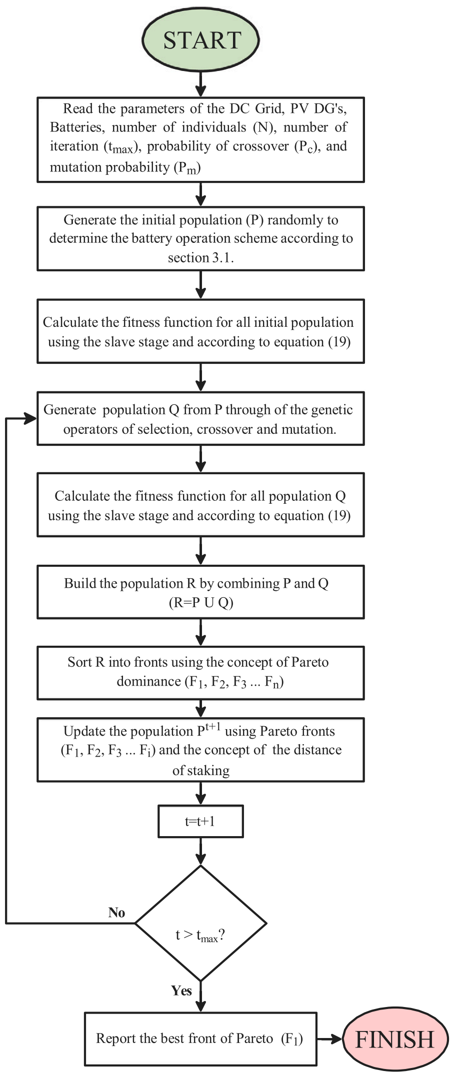

Fig. 5. Master stage proposed based on non-dominated sorting genetic algorithm.

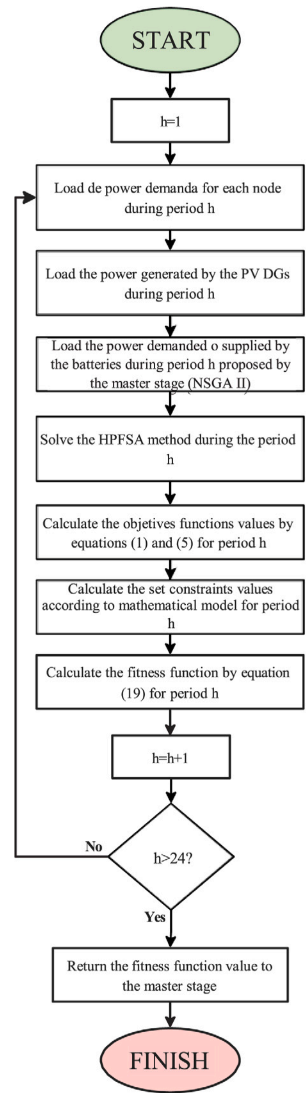

Fig. 6. Slave stage proposed using the hourly power flow based on successive approximation method.

3.3.2. Slave stage: hourly power flow based on successive approximations method

For the second part of the methodology, a slave stage based on an hourly load flow method using the successive approximations method is proposed. This method can be consulted in [42]. This method was selected due to its high efficiency, fast convergence in distribution systems, and ease of implementation in computational tools, as it does not require derivatives in its formulation. In comparison with more complex methodologies for optimal power flow determination, such as those presented in [45,46], the successive approximations method achieves the same results in terms of power losses and does so within a similar computation time [47]. Therefore, the successive approximations method is well-suited for distribution system analysis, as it provides both accuracy and efficiency without the additional computational load that more sophisticated methods may entail.

The HPFSA is performed for each hourly interval during a typical operating day, for which it is necessary to know the generation data of the existing distributed resources in the network (i.e. ESS and PV generators), as well as the hourly demand of the entire DC distribution network. In this way, it is possible to perform an analysis of the effect of integrating the ESS into the grid, in this case the effect is calculated by evaluating the two objective functions described in Eqs. (1) and (5). It is important to highlight that in this research it is considered that the PV generators are in operation at the maximum power point.

The HPFSA method then makes it possible to determine, in each analysis period, the voltage profiles of all the nodes of the system, the loadability of the lines and the power flows. It is thus possible to evaluate all the operational constraints of the network according to the mathematical model proposed in Section 2.2. To allow further exploration of the solution space of the problem addressed in this article, the adaptation function proposed in Eq. (19) is evaluated and used as a metric of the quality of each of the ESS operation configurations proposed by the master stage. It is worth noting that all proposed solutions are evaluated on their two objective functions, and if any of them violate the operating conditions, they are penalized, thus allowing the search process to advance beyond infeasible regions and thus further explore the search space. Fig. 6 shows the steps of the proposed slave stage.

4. Test system and considerations

In this paper, as a test system, it has been used a DC grid that considers the variable power generation and demand of the city of Medellín-Colombia [30]. This DC grid is composed by 33 buses and 32 branches, considering as base values 12.66 kV and 100 kVA. The

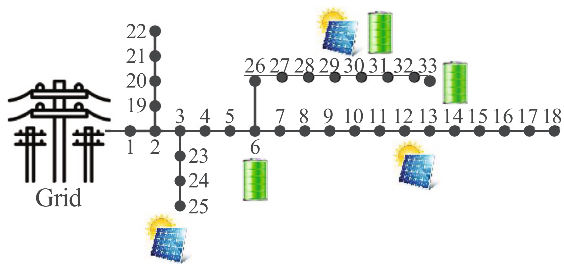

Fig. 7. Electrical diagram of 33 bus test system DC grid [30].

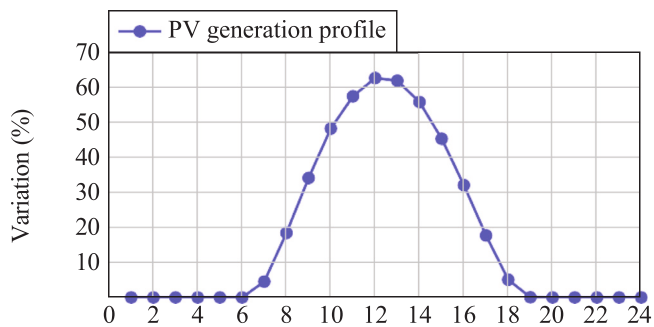

Fig. 8. Daily PV generation curves for an operation day from Medellín-Colombia.

electrical diagram that describes this test system is presented below (see Fig. 7).

As it can be seen in the last figure, the DC grid considered in this document consists of three distributed generators installed at bus 12, 25, and 30, with a nominal power of 1125 kW, 1320 kW, and 999 kW, respectively. The behavior of the PV generation for the region of Medellín-Colombia is shown in Fig. 8. These data were obtained using the irradiance and ambient temperature values reported by the NASA database for Medellín-Colombia, and using the parameters of polycrystalline PV modules [48].

The 33-bus DC grid analyzed here contains three integrated ion lithium ESS on buses 6, 14, and 31, respectively. With rated power of 2000 kW, 1500 kW and 1000 kW. By presenting the ESS installed on buses 14 and 31 charge and discharge times of 4 h, while the ESS located on bus 6 presents a 5 h for both times. It is important to highlight that in order to obtain the best performance of the ESS, in this work it is considered that the ESS start and end the day with a capacity of 50% of SOC, with a minimum and maximum SOC value of 10% and 90%.

Table 1 describes the parameters of the DC grids used. This table shows, from left to right, the number of lines, the sending and receiving buses, the resistances and reactances of the lines, the power required at the receiving bus, and the maximum current level of each line. The power demand behavior of the grid users is illustrated in Fig. 9, taken from the data reported for the utility of Medellín city.

Given that the voltage at the substation node is 1 p.u. as an operational consideration, this document adopts a maximum voltage profile variation of ±10% has been adopted [42]. Furthermore, in all the periods studied, the lines that constitutes the DC grid must operate under the maximum current limit assigned (see last column of the Table 1).

5. Simulation results

In this section, the simulation results obtained after applying the proposed methodology and the comparison methodologies used to solve the ESS power coordination problem in DC grids are presented. This paper employs two comparison methodologies: MOPSO and

Table 1
Electrical parameters considered for the DC grids of 33 buses.

Line l	Bus i	Bus j	R_{ij} (Ω)	P_j (kW)	$I_{ij}^{\max}$ (A)
1	1	2	0.0922	100	385
2	2	3	0.4930	90	355
3	3	4	0.3660	120	240
4	4	5	0.3811	60	240
5	5	6	0.8190	60	240
6	6	7	0.1872	200	110
7	7	8	1.7114	200	85
8	8	9	1.0300	60	70
9	9	10	1.0400	60	70
10	10	11	0.1966	45	55
11	11	12	0.3744	60	55
12	12	13	1.4680	60	55
13	13	14	0.5416	120	40
14	14	15	0.5910	60	25
15	15	16	0.7463	60	20
16	16	17	1.2890	60	20
17	17	18	0.7320	90	20
18	2	19	0.1640	90	40
19	19	20	1.5042	90	25
20	20	21	0.4095	90	20
21	21	22	0.7089	90	20
22	3	23	0.4512	90	85
23	23	24	0.8980	420	85
24	24	25	0.8960	420	40
25	6	26	0.2030	60	125
26	26	27	0.2842	60	110
27	27	28	1.0590	60	110
28	28	29	0.8042	120	110
29	29	30	0.5075	200	95
30	30	31	0.9744	150	55
31	31	32	0.3105	210	30
32	32	33	0.3410	60	20

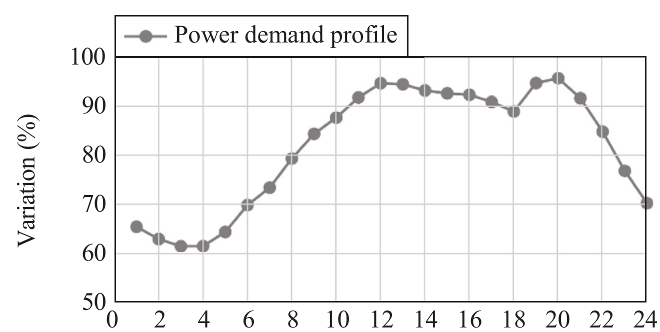

Fig. 9. Daily PV generation curves for Medellín-Colombia.

MOALO. These methodologies were implemented because they have been widely used in the specialized literature, as evidenced by the following articles [49–54]. These documents highlight the efficiency, capacity to handle conflicting objectives, and the flexibility and adaptability of each algorithm. All simulations were conducted using MATLAB software, version R2022a, on a Dell Precision 3450 workstation equipped with an Intel(R) Core(TM) i9-11900 CPU running at 2.50 GHz, with 8 cores and 64.0 GB of RAM, operating on the 64-bit Windows 10 Pro operating system.

To ensure a fair comparison between the methodologies used for comparison purposes and the methodology proposed in this document, it is first necessary to define the input parameters of each algorithm. The input parameters have been selected using PSO. This stage is crucial since each algorithm has its own formulation with various parameters to explore and exploit the solution space. Tuning identifies the most suitable values; however, manually adjusting these parameters is tedious and time-consuming [55]. To automate this, the PSO was used in a preliminary tuning stage.

PSO starts by creating an initial population representing the parameters of the algorithms used in this paper (shown in Table 2). This

Table 2
Input parameters for the algorithms used in this study.

Algorithm	Parameter	Value
MOPSO	Number of individuals (N_i)	93
	Number of iterations (t_{max})	981
	Cognitive component (C_1)	1.0507
	Social component (C_2)	1.5382
	Maximum inertia (W_{max})	0.6875
	Minimum inertia (W_{min})	0.0889
	Number of grids per dimension (N_{grid})	45
	Repository size (N_{rep})	100
MOALO	Number of individuals (N_i)	100
	Number of iterations (t_{max})	1000
	Repository size (N_{rep})	100
NSGAII	Number of individuals (N_i)	99
	Number of iterations (t_{max})	1000
	Probability of crossover (P_c)	0.6058
	Mutation probability (P_m)	0.0456
	Mutation intensity (M_s)	0.0829
	Repository size (N_{rep})	99

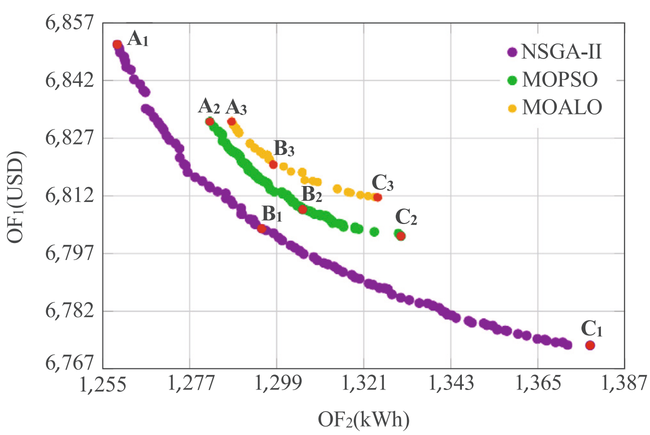

Fig. 10. Best Pareto fronts obtained by the methods used.

population is then used as input parameters for the multi-objective algorithms in the master–slave methodology. The main objective is to determine the value of both objective functions and estimate the parameter quality.

Once the parameter quality is determined, the iterative process continues, guiding PSO towards the most promising parameter combination. By the end of this process, PSO finds the parameters that best fit the optimization problem for the multi-objective algorithms in the master–slave methodology.

For this particular case, the algorithm uses 40 individuals, a maximum number of iterations of 300, acceleration coefficients C_1 and C_2 of 2, and maximum and minimum inertia W_{max} and W_{min} of 0.9 and 0.1, respectively. The resulting input parameters are listed in the Table 2.

5.1. Pareto front analysis

Fig. 10 presents the best Pareto fronts obtained through the methodologies used to solve the ESS power coordination problem in DC grids.

In this figure, each point on the Pareto front symbolizes a feasible solution that satisfies all the constraints of the mathematical model. In this context, it can be observed that the three methods show a similar behavior. As one objective function is minimized, the other tends to increase, confirming that for the ESS power coordination problem in DC grids, the objective functions of cost and energy losses are in conflict.

This behavior can be explained as follows: As PV generators/ESS inject power, they cover a percentage of the power demand at the node where these devices are located. If PV generators/ESS inject more power than the node's demand, reverse power flow is generated, i.e., from the demand node to the rest of the nodes, redistributing the system's power flows. Consequently, the excess power helps cover part of the demand in the rest of the system, reducing dependence on the substation node and lowering system costs. However, if too much power is injected into the system, the circulating current can increase to the point of reaching the thermal limits of the distribution lines, increasing system energy losses but reducing system costs.

On the other hand, if PV generators/ESSs power injection is reduced to the point that it only covers the node's demand, the circulating current in the system decreases, reducing energy losses. However, dependence on the substation node increases, as it must generate more power to cover the remaining demand, thus increasing system costs. Similarly, it should be noted that ESS will absorb power during periods when energy prices are lower and supply energy when prices are higher. Likewise, PV generators will supply energy throughout the solar window (from 6:00 to 19:00), reaching the peak generation at 12:00. These two considerations can affect power injection by PV generators/ESS throughout the day.

However, it is observed that the proposed method outperforms MOALO and MOPSO in terms of dominance, diversity, and cardinality. The solutions obtained with NSGAII dominate those found by MOALO and MOPSO, showing equal or superior performance in both objective functions. In addition, NSGAII offers a greater number of solutions, providing network operators with a wide range of options from which to choose the solution that best suits their specific needs. Finally, by providing a greater number of solutions, NSGAII achieves a more comprehensive exploration of the solution space, resulting in a better approximation of the true Pareto front.

5.2. Methodology for selecting the best solution

Since there is no single best solution on the Pareto front — meaning that in multi-objective optimization, it is not possible to select one best solution — this document has chosen to study three points of interest: Point A, Point B, and Point C. These points correspond to the extremes and the inflection point of the Pareto front. It is also important to note that subscript 1 is associated with NSGAII, subscript 2 with MOPSO, and subscript 3 with MOALO. In this convention, points A represent the minimum losses found by each method, points B are the inflection points of each Pareto front, where the solution that allows the maximum simultaneous reductions of both objective functions is located, i.e., it represents the solution in which one objective function cannot be improved without worsening the other. Finally, point C represents the lowest operating costs found by each method.

Point B is considered the optimal point on the Pareto front. There are various methodologies for selecting this inflection point, such as the one presented in [56], which is based on fuzzy logic, where the membership values of each point on the Pareto front depend on the decision-maker's criteria. Other methodologies are presented in [57], which include three methods: the Method based on specific restrictions, which relies on the decision-maker's experience; the Analytical Method, which analyzes variations in conflicting objective functions to identify optimal solutions; and the Quantitative Method, which minimizes the normalized sum of the objective functions, allowing for a balanced point between both objectives.

In our document, we chose the quantitative method for selecting the compromise point on the Pareto front due to its objectivity and its ability to accurately balance cost minimization and energy loss objectives. This approach avoids the subjectivity inherent in the method based on specific restrictions, which relies on predetermined objective values, and surpasses the limitations of the analytical method, which offers local solutions without jointly optimizing both objectives. Additionally, we dismissed the fuzzy logic-based method, as it introduces uncertainty into the preferences and generates less objective solutions. In contrast,

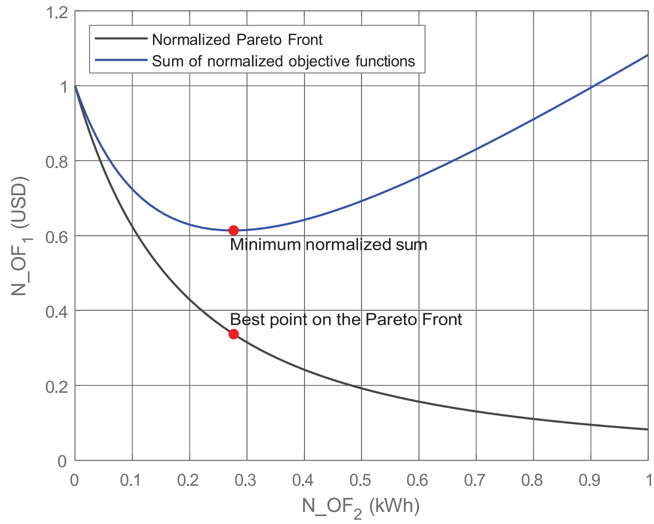

Fig. 11. Graphical representation of the best point selection on the Pareto front. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3
Simulation results for the different points of the Pareto Front of the 33-node test system.

Methodology	Ecosts (USD)	Energy loss (kWh)
Point A		
Methodology	Ecosts (USD)	Energy loss (kWh)
Base case	6865.0130	1357.8724
MOALO	6831.3875	1287.6348
MOPSO	6831.3586	1282.1532
NSGAII	6851.2594	1258.7924
Point B		
Methodology	Ecosts (USD)	Energy loss (kWh)
Base case	6865.0130	1357.8724
MOALO	6820.1180	1298.1673
MOPSO	6808.4824	1305.5722
NSGAII	6803.5274	1295.2462
Point C		
Methodology	Ecosts (USD)	Energy loss (kWh)
Base case	6865.0130	1357.8724
MOALO	6811.6183	1324.5104
MOPSO	6801.5734	1330.3520
NSGAII	6773.1551	1378.2148

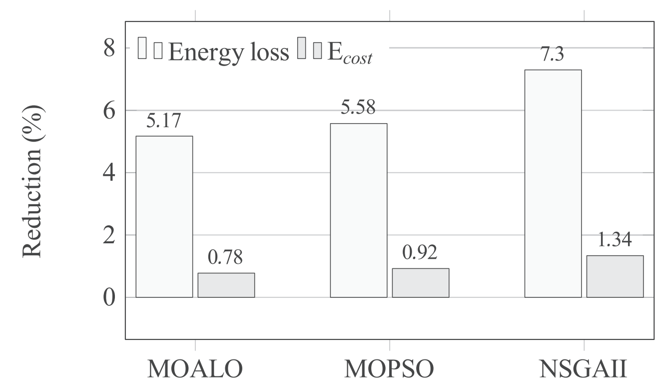

Fig. 12. The reductions achieved by the optimization methodologies in the 33-node test system compared to the base case at points A and C of the Pareto Front.

the quantitative method allows for the direct and robust identification of a globally optimal solution, well-suited to our multi-objective optimization needs.

The quantitative method involves the following steps [57]:

1. The values of operating costs and energy losses are normalized to facilitate comparison. This normalization calculates the relative values of each point on the Pareto front with respect to the maximum values found for each objective function, as shown by the black curve in Fig. 11.
2. The normalized values of the objective functions are summed for each point on the Pareto front, as illustrated by the blue curve in Fig. 11.
3. The point with the minimum normalized sum of objectives is selected (indicated by the red point in Fig. 11). This point, reflected on the Pareto Front, represents the balance between both objective functions. In other words, it is the point where costs cannot be improved without worsening losses, and vice versa (i.e., Point B).

This process ensures a robust and objective selection of the optimal point on the Pareto front, considering the specific characteristics of the multi-objective problem.

5.3. Main simulation results analysis

In this way, Table 3 shows the simulation results obtained in the 33-node test system corresponding to points A, B, and C in Fig. 10. This table displays the following information: the solution method that was employed, the network operating costs in USD and the network's energy losses in kWh. It is important to note that this table is divided into three sections. The first section shows the results obtained at point A, the second section shows the results obtained at point B, and the third section shows the results obtained at point C.

The results of the simulations performed on the 33-node test system show the following: at point A, corresponding to the minimum energy losses scenario, all methodologies achieve a reduction of more than 70 kWh compared to the base case. However, as shown in the table, NSGAII stands out as the methodology with the best results in terms of response quality. This algorithm achieves reductions of 99.0800 kWh compared to the base case, 28.8423 kWh compared to MOALO, and 23.3607 kWh compared to MOPSO.

At the other end of the Pareto front (point C), in the minimum operating cost scenario, all three methodologies achieve savings of more than 53 USD compared to the base case. In this scenario, NSGAII continues to stand out as the methodology with the best results, with reductions of 91.8579 USD compared to the base case, 38.4632 USD compared to MOALO, and 28.4183 USD compared to MOPSO.

Finally, at the inflection point of the Pareto front (point B), the three methodologies analyzed allow a reduction of more than 44 USD in operating costs and 59 kWh in energy losses compared to the base case. Once again, it is confirmed that the proposed methodology achieves the best results. In this case, reductions of 61.4857 USD and 62.6263 kWh are obtained compared to the base case, 16.5907 USD and 2.9212 kWh compared to MOALO, and 4.9550 USD and 10.3261 kWh compared to MOPSO. The above shows that the NSGAII consistently achieves the most significant reductions in both energy losses and operating costs compared to the base case and other methodologies (MOALO and MOPSO). This suggests its effectiveness in optimizing the 33-node test system with both objectives in mind simultaneously.

In addition, Figs. 12 and 13 show the percentage reductions achieved by each optimization method compared to the baseline case. Fig. 12 shows the reduction in energy losses and operating costs for points A and C of the Pareto front, respectively. On the other hand, Fig. 13 shows the reduction in both energy losses and operating costs for the inflection point of the Pareto front.

According to Fig. 12, it can be observed that all three methodologies applied achieve reductions greater than 5.10% in energy losses at point

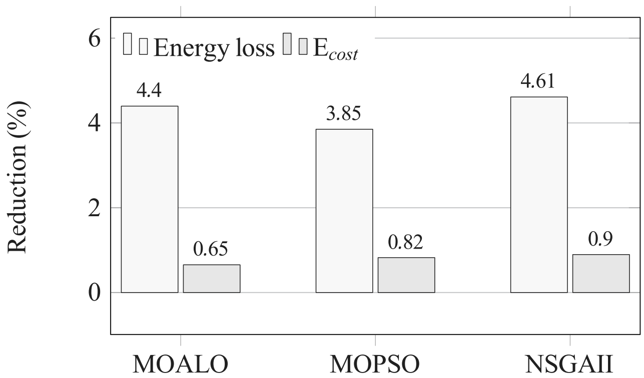

Fig. 13. The reductions achieved by the optimization methodologies in the 33-node test system compared to the base case at inflection point of the Pareto Front.

Table 4
Performance of the methods used around the inflection point.

Operating cost				
Methodology	Best (USD)	Average (USD)	STD (%)	Time (s)
MOALO	6793.0884	6806.9781	0.0784	201.4363
MOPSO	6776.2520	6797.9576	0.1401	33.6964
NSGAII	6773.9027	6793.2702	0.0677	89.6648
Energy losses				
Methodology	Best (USD)	Average (USD)	STD (%)	Time (s)
MOALO	1296.3247	1314.2221	0.6706	201.4363
MOPSO	1293.1732	1326.6055	1.1765	33.6964
NSGAII	1282.8240	1304.0030	0.5705	89.6648

A and reductions greater than 0.70% in operating costs at point C. It is evident that the proposed NSGAII allows the highest reductions in both scenarios, reducing energy losses at point A by 7.2967% compared to the base case and operating costs at point C by 1.3381% compared to the base case. Furthermore, upon analyzing the reductions achieved at the inflection point (Fig. 13), it is observed that the proposed methodology achieves the greatest reductions compared to the other solution methodologies. In this case, NSGAII allows for simultaneous reductions of 4.6121% in energy losses and 0.8956% in operating costs compared to the base case. The analysis strongly suggests that NSGAII is a promising methodology for optimizing distribution systems, offering significant improvements in both energy efficiency and cost savings.

Moreover, in this study, 100 consecutive evaluations of the proposed methodology were carried out to determine its efficiency and robustness in solving the ESS power coordination problem in DC grids. To assess the efficiency and robustness of NSGAII, repeatability at the inflection point in the 100 resulting Pareto Fronts was analyzed, as well as the average computation time required for the proposed methodology to converge to the optimal Pareto Front. The results obtained are displayed in Table 4. This table contains the methodology used, the best result achieved, the average result, the percentage standard deviation, and the average computation time for the two evaluated objective functions.

According to the results presented in Table 4, it is observed that NSGAII achieves superior results compared to MOALO and MOPSO in solving the ESS power coordination problem in DC grids. In terms of the best response for the inflection point, NSGAII achieves a value of 6773.9027 USD, which is an improvement of 19.1053 USD over the best response of MOALO and 2.9212 USD over the best response of MOPSO. Regarding energy losses, NSGAII achieves a response of 1282.8240 kWh for the inflection point, representing an improvement of 13.5007 kWh compared to MOALO and 10.3492 kWh compared to MOPSO. The application of NSGAII enable the network operator to find an optimal equilibrium point that simultaneously ensures the lowest energy losses and operating costs in DC grids.

Regarding the average response, the results obtained indicate that NSGAII achieves the best results for both operating costs and energy losses compared to the other methodologies used. For the operating costs, the proposed methodology obtains an average value of 6793.2702 USD. NSGAII shows an improvement of 13.7079 USD over MOALO and an improvement of 4.6874 USD over MOPSO. In the case of energy losses, the proposed methodology achieves an average value of 1304.0030 kWh, which is an improvement of 10.2191 kWh over MOALO and an improvement of 22.6025 kWh over MOPSO. Therefore, on average, NSGAII is the best option for optimizing both operating costs and energy losses.

Similarly, from the data presented in Table 4, it can be seen that NSGAII is the solution methodology that achieves the best results in terms of standard deviation. For operating costs, NSGAII achieves a standard deviation of 0.0677%, which is approximately 13.6480% lower than that of MOALO and 51.6773% lower than that of MOPSO. Similarly, for energy losses, the proposed methodology achieves a standard deviation of 0.5705%, which is approximately 14.9269% lower than that of MOALO and 51.5087% lower than that of MOPSO. Even though the solutions of the best Pareto Front obtained by NSGAII occupy a broader search space compared to MOALO and MOPSO, i.e., the results are more dispersed, their standard deviation is lower than that of these methodologies. These findings suggest that every time NSGAII is executed to solve this problem, the optimal Pareto front or a front with very close non-dominated solutions will be found.

Finally, in terms of average computation times, the best methodology turned out to be MOPSO. On average, MOPSO takes approximately 33.6964 s to establish a Pareto Front with non-dominated solutions. On the other hand, NSGAII requires 89.6648 s. These processing times can be considered short given the complex nature of the NLP model involving continuous variables (i.e., a solution space with infinite combinations). The proposed methodology requires less than a minute and a half to converge to an optimal Pareto Front. Thus, NSGAII is a methodology that offers network operators the ability to find a solution that simultaneously minimizes operating costs and energy losses with low computational cost.

On the other hand, the ESS operation scheme proposed by NSGAII is shown in Fig. 14. From this figure, it can be observed that the solutions provided by the proposed methodology comply with the constraints established in the mathematical model. In this regard, it can be observed that the SoC of the ESS fluctuates in the range of 10% to 90%, and furthermore, both the beginning and the end of the SoC are located at 50%.

In Fig. 14a it can be seen that during the first 6 h ESS 1 and 3 are partially charged, while ESS 2 is discharged below 20%. From time interval 8, when the PV generators supply more than 20% of their nominal capacity to the grid, all three ESS start to charge until they reach their maximum charge capacity at hour 13. Then, from hour 16, when the PV generators reduce their power injection below 30% of their rated capacity, all three ESS discharge until they reach their final state of charge, supporting the system operation as the loads require more than 90% of their rated capacity. In the case of minimizing energy losses, the operation of the ESS is directly linked to the power injection of the PV generators. During periods when the PV generators are injecting power, the ESS tend to charge, while during periods when the generators are not injecting power, the ESS tend to discharge to support system operation. In this way, energy transfer is facilitated and energy losses are reduced.

In addition, Fig. 14b shows the opposite end of the Pareto Front provided by NSGAII, specifically the operating scheme that manages to reduce the operating costs of the system. During the first 5 time intervals, the three ESS are charged to their maximum capacity, taking advantage of the fact that the energy price is below 80%. From the 6th hour, when the energy price rises above 85%, the three ESS are partially discharged in order to economically meet the system's power

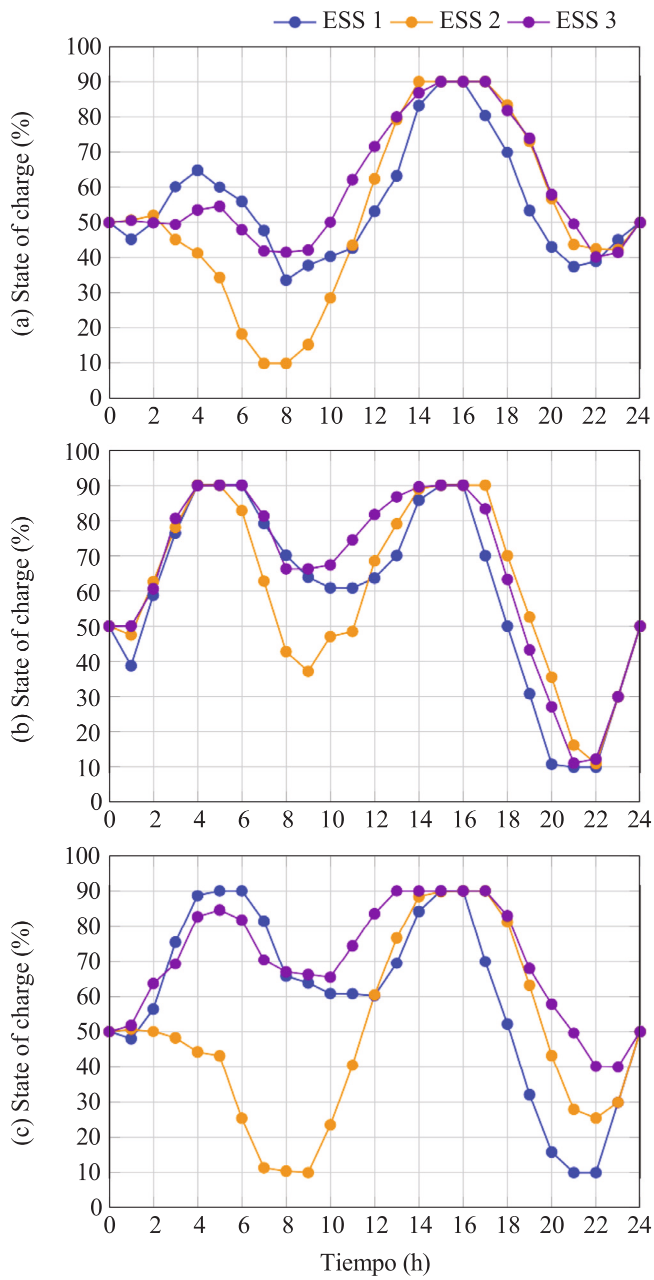

Fig. 14. ESS operation scheme provided by NSGAII: (a) minimum energy losses, (b) minimum lower operating costs, and (c) inflection point.

demand, which is above 70% of their nominal capacity. Then, around noon, the ESS are recharged to start a discharge process from hour 17. The ESS supply all their energy to the system when the energy price and the power demand are the highest of the day, exceeding 95% in both cases, thus ensuring the economic operation of the system. As can be seen, the operating scheme of the ESS is determined by the variable cost of energy, as they try to charge when the cost of energy is lowest and discharge when the price of energy is higher.

Finally, Fig. 14c shows the ESS operation scheme at the inflection point of the best Pareto Front provided by NSGAII. During the first 6 time intervals, ESS 1 and 3 are charged to more than 80% of their capacity, while ESS 2 is completely discharged. ESS 1 and 3 are charged economically, while ESS 2 injects power to cover part of the demand. From hour 7, ESS 1 and 3 undergo discharging actions as the energy price exceeds 90%, reducing the energy purchase at the substation node. Then, in time interval 9, all three ESS are charged until they reach their maximum state of charge, taking advantage of the fact that the photovoltaic generators are injecting power above 35% of their nominal capacity. Then, from hour 17 to hour 22, the three ESS are partially discharged, since the power injected by the PV generators is less than 20% of their nominal capacity, and furthermore, the energy price and the power demand are above 95%, leading to an intelligent and economical ESS operation. From this figure, it can be seen that the operation scheme is determined by the variable energy price and the availability of the solar resource. To achieve both economic and efficient operation, ESS prioritize charging during periods of low energy prices or when PV generators are supplying power to the grid. Similarly, ESS tend to discharge during periods of high energy prices or when generators are not supplying power to the system.

Additionally, in Fig. 15, the profiles of maximum and minimum voltage, as well as the maximum loadability on the distribution lines obtained through NSGAII, are presented.

In Fig. 15a, the maximum and minimum voltage levels for the different nodes of the test system are presented. It can be observed that the voltage levels for each node remain within the allowed limits, i.e., they oscillate between $\pm 10\%$ of the nominal voltage. In this specific case, the maximum voltage is always found at the substation node with a value of 1 p.u., while the minimum voltage is recorded at node 18 with a value of 0.9285 p.u., which corresponds to the terminal node of the test system, i.e., the node farthest from the substation.

On the other hand, in Fig. 15b, the maximum load capacity level for the different distribution lines that make up the test system is presented. It is observed that the maximum load capacity is found in lines 13 and 30, with a value very close to 100%. This was expected, given that they are the lines closest to generators FV 2 and 3, as well as to ESS 2 and 3. This demonstrates that the solutions obtained by NSGAII satisfy the constraints imposed in the mathematical model.

5.4. Managerial implications

The findings of this study offer several valuable insights for managers and operators of DC distribution networks, especially concerning the efficient coordination of ESS. By implementing the proposed methodology, distribution system managers can optimize the operation of ESS, achieving a balance between operational cost and energy losses. This balance is critical in contexts where fluctuations in demand and renewable generation create instability risks, making it essential to maintain system reliability while minimizing operational expenses.

The methodological approach using a master–slave structure not only ensures fast and reliable convergence but also minimizes computational costs, making it suitable for real-time or near-real-time applications. This efficiency can lead to significant cost and energy savings in operational management, as it allows for quick adjustments to ESS operations in response to dynamic system conditions without requiring extensive computational resources.

Additionally, the model provides decision-makers with a framework for scenario planning and long-term strategic decisions. By simulating and analyzing different ESS configurations, managers can better understand the trade-offs between energy costs and network losses. This understanding is crucial for designing resilient distribution systems that can accommodate higher levels of renewable energy penetration and adapt to evolving regulatory requirements for energy efficiency and sustainability. The proposed methodology enables managers and operators to enhance both the financial and technical performance of DC distribution networks, supporting a more sustainable and resilient energy infrastructure.

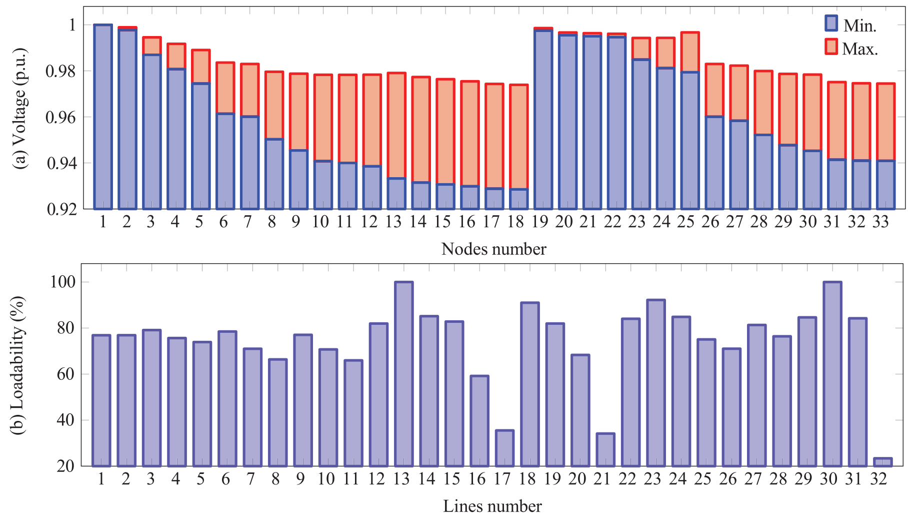

Fig. 15. Behavior of the electrical variables of the test system after implementation of NSGAII: (a) maximum and minimum voltage at the nodes, and (b) maximum loadability of the distribution lines.

6. Conclusions and future work

This study presents an optimized approach for coordinating ESS in DC distribution networks with high solar PV penetration and variable demand, aiming to reduce both operating costs and energy losses. Using a master–slave methodology based on NSGA-II, this approach determines the optimal ESS dispatch scheme and evaluates objective functions via hourly power flow analysis. Results were validated through comparison with MOPSO and MOALO algorithms, highlighting the effectiveness of the proposed method.

6.1. Main results

The proposed methodology significantly enhances the performance of the test system over the base case scenario. Specifically, it achieved reductions of over 91.8579 USD (1.3381%) in operational costs and 99.0800 kWh (7.2967%) in energy losses at the extremes of the Pareto Front. At the inflection point of the Pareto Front, the NSGA-II algorithm achieved minimal operational costs and energy losses, with values of 6803.5274 USD and 1295.2462 kWh, respectively, underscoring the economic and technical benefits that effective ESS coordination can provide to grid operators. Additionally, NSGA-II exhibited superior performance in terms of optimal solutions, average response, and low standard deviation, making it the most reliable option for optimizing operational costs and energy losses within the system. The proposed approach shows potential for adaptation to other types of distribution networks, including AC networks and hybrid AC/DC configurations, as the ESS coordination methodology and multi-objective optimization framework can be adjusted for different network parameters and generation profiles. Furthermore, the methodology is flexible enough to incorporate various types of distributed generation sources, such as wind turbines or microgrids, broadening its applicability to a wider range of energy distribution scenarios.

6.2. Limitations

Despite its advantages, the application of multi-objective metaheuristic algorithms (e.g., MOPSO, MOALO, and NSGA-II) to simultaneously optimize costs and energy losses through ESS management has certain limitations. Achieving a balanced optimization of conflicting objectives can be challenging, and these algorithms often require extensive parameter tuning and computational resources to thoroughly explore the solution space. Additionally, the algorithms' performance may decrease with the complexity of the network or under dynamic, real-time operational scenarios, potentially leading to suboptimal solutions or longer convergence times.

6.3. Future works

To extend the impact and scope of this study, several future research directions are proposed:

- Expanding the mathematical model to incorporate bipolar DC grids with unbalanced loads would enhance the model's applicability and provide insights into more diverse network conditions.
- Adding an environmental objective function to minimize CO_2 emissions from conventional energy sources would support a more sustainable operation framework, addressing both economic and ecological objectives in ESS coordination.
- Dynamic Optimization of ESS Sizing and Dispatch: Including ESS sizing as an optimization variable alongside ESS and distributed generator dispatch could enhance the adaptability and economic feasibility of the solution, providing operators with a more holistic energy management tool.
- Incorporating solar PV dispatch into the model would allow for improved power management within the DC network, enhancing both technical and economic indicators. This would also enable the system to respond more effectively to fluctuations in renewable generation.
- Developing real-time optimization capabilities would allow the model to handle dynamic conditions and respond to real-time changes in demand and generation. Future research should explore adaptive algorithms capable of managing ESS coordination efficiently in continuously changing operational scenarios.

These directions aim to refine and expand the model's application, contributing to a more versatile and robust framework for managing ESS in active DC distribution networks.

CRediT authorship contribution statement

Brandon Cortés-Caicedo: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Luis Fernando Grisales-Noreña:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Oscar Danilo Montoya:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Rubén Iván Bolaños:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Javier Muñoz:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The author, certify that all authors of this paper participated in the development of the conceptualization, methodology, software, formal analysis, investigation, data curation, writing, visualization, project administration and funding acquisition.

All we affirm that have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version. In addition, we affirm that our manuscript has not been submitted to, nor is under review at, another journal or other publishing venue. Lastly, we affirm that we do not have any conflicts of interest.

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Data availability

No data was used for the research described in the article.

References

[1] J. Rodriguez, F. Blaabjerg, M.P. Kazmierkowski, Energy transition technology: The role of power electronics, Proc. IEEE 111 (4) (2023) 329–334, http://dx.doi.org/10.1109/jproc.2023.3257421.

[2] M. Kashyap, S. Kansal, R. Verma, Sizing and allocation of DGs in a passive distribution network under various loading scenarios, Electr. Power Syst. Res. 209 (2022) 108046, http://dx.doi.org/10.1016/j.epsr.2022.108046.

[3] S. Lakshmi, S. Ganguly, Transition of power distribution system planning from passive to active networks: A state-of-the-art review and a new proposal, in: Green Energy and Technology, Springer, Singapore, 2017, pp. 87–117, http://dx.doi.org/10.1007/978-981-10-7188-1_4.

[4] J. Caballero-Peña, C. Cadena-Zarate, A. Parrado-Duque, G. Osma-Pinto, Distributed energy resources on distribution networks: A systematic review of modelling, simulation, metrics, and impacts, Int. J. Electr. Power Energy Syst. 138 (2022) 107900, http://dx.doi.org/10.1016/j.ijepes.2021.107900.

[5] K. Twaisan, N. Barışçı, Integrated distributed energy resources (DER) and microgrids: Modeling and optimization of DERs, Electronics 11 (18) (2022) 2816, http://dx.doi.org/10.3390/electronics11182816.

[6] O.D. Montoya, W. Gil-González, Dynamic active and reactive power compensation in distribution networks with batteries: A day-ahead economic dispatch approach, Comput. Electr. Eng. 85 (2020) 106710, http://dx.doi.org/10.1016/j.compeleceng.2020.106710.

[7] O.D. Montoya, F.M. Serra, C.H. De Angelo, On the efficiency in electrical networks with AC and DC operation technologies: A comparative study at the distribution stage, Electronics 9 (9) (2020) 1352, http://dx.doi.org/10.3390/electronics9091352.

[8] A. Garces, Uniqueness of the power flow solutions in low voltage direct current grids, Electr. Power Syst. Res. 151 (2017) 149–153, http://dx.doi.org/10.1016/j.epsr.2017.05.031.

[9] V. Azbe, R. Mihalic, Distributed generation from renewable sources in an isolated DC network, Renew. Energy 31 (14) (2006) 2370–2384, http://dx.doi.org/10.1016/j.renene.2006.01.001.

[10] H.E. Gelani, F. Dastgeer, M. Nasir, S. Khan, J.M. Guerrero, AC vs. DC distribution efficiency: Are we on the right path? Energies 14 (13) (2021) 4039, http://dx.doi.org/10.3390/en14134039.

[11] H. Gelani, F. Dastgeer, K. Siraj, M. Nasir, K. Niazi, Y. Yang, Efficiency comparison of AC and DC distribution networks for modern residential localities, Appl. Sci. 9 (3) (2019) 582, http://dx.doi.org/10.3390/app9030582.

[12] O. Unigwe, D. Okekunle, A. Kiprakis, Smart coordination of battery energy storage systems for voltage control in distribution networks with high penetration of photovoltaics, J. Eng. 2019 (18) (2019) 4738–4742, http://dx.doi.org/10.1049/joe.2018.9286.

[13] M. Ahmadi, O.B. Adewuyi, M.S.S. Danish, P. Mandal, A. Yona, T. Senjyu, Optimum coordination of centralized and distributed renewable power generation incorporating battery storage system into the electric distribution network, Int. J. Electr. Power Energy Syst. 125 (2021) 106458, http://dx.doi.org/10.1016/j.ijepes.2020.106458.

[14] L.F. Grisales-Noreña, O.D. Montoya, J.C. Hernández, An efficient EMS for BESS in monopolar DC networks with high penetration of renewable generation: A convex approximation, Batteries 9 (2) (2023) 84, http://dx.doi.org/10.3390/batteries9020084.

[15] H. Song, M. Gu, C. Liu, A.M. Amani, M. Jalili, L. Meegahapola, X. Yu, G. Dickeson, Multi-objective battery energy storage optimization for virtual power plant applications, Appl. Energy 352 (2023) 121860, http://dx.doi.org/10.1016/j.apenergy.2023.121860.

[16] H.A. Taha, M.H. Alham, H.K.M. Youssef, Multi-objective optimization for optimal allocation and coordination of wind and solar DGs, BESSs and capacitors in presence of demand response, IEEE Access 10 (2022) 16225–16241, http://dx.doi.org/10.1109/access.2022.3149135.

[17] J. Liu, H. Wang, Y. Du, Y. Lu, Z. Wang, Multi-objective optimal peak load shaving strategy using coordinated scheduling of EVs and BESS with adoption of MORBHPSO, J. Energy Storage 64 (2023) 107121, http://dx.doi.org/10.1016/j.est.2023.107121.

[18] R. Manojkumar, C. Kumar, S. Ganguly, H.B. Gooi, S. Mekhilef, J.P.S. Catalão, Rule-based peak shaving using master-slave level optimization in a diesel generator supplied microgrid, IEEE Trans. Power Syst. 38 (3) (2023) 2177–2188, http://dx.doi.org/10.1109/tpwrs.2022.3187069.

[19] B. Cortés-Caicedo, J. Ocampo-Toro, R.I. Bolaños, O.D. Montoya, L.F. Grisales-Noreña, A multi-objective PSO for DC microgrids: Efficient battery management to minimize energy losses and operating costs, J. Energy Storage 96 (2024) 112550, http://dx.doi.org/10.1016/j.est.2024.112550.

[20] J. Montano, J.P. Guzmán, O.D. Garzón, A.M.R. Barrera, Techno-economic approach for energy management system: Multi-objective optimization algorithms for energy storage in standalone and grid-connected DC microgrids, J. Energy Storage 102 (2024) 114069.

[21] O. Aldosari, Z.M. Ali, S.H. Abdel Aleem, M.H. Mostafa, Optimizing microgrid performance: Strategic integration of electric vehicle charging with renewable energy and storage systems for total operation cost and emissions minimization, PLoS One 19 (10) (2024) e0307810.

[22] F.A. Kassab, B. Celik, F. Locment, M. Sechilariu, S. Liaquat, T.M. Hansen, Optimal sizing and energy management of a microgrid: A joint MILP approach for minimization of energy cost and carbon emission, Renew. Energy 224 (2024) 120186.

[23] O.D. Montoya, F.M. Serra, W. Gil-González, L.F. Grisales-Noreña, J.C. Hernández, Multi-objective dispatch for batteries and renewable generation sources in distribution grids using a weighting-based convex approach while considering uncertainty, IEEE Access (2024).

[24] S. Iqbal, M. Nasir, M.F. Zia, K. Riaz, H. Sajjad, H.A. Khan, A novel approach for system loss minimization in a peer-to-peer energy sharing community DC microgrid, Int. J. Electr. Power Energy Syst. 129 (2021) 106775, http://dx.doi.org/10.1016/j.ijepes.2021.106775.

[25] J.-S. Lee, Y.-J. Choi, A stability improvement method of DC microgrid system using passive damping and proportional-resonance (PR) control, Sustainability 13 (17) (2021) 9542, http://dx.doi.org/10.3390/su13179542.

[26] J. Sun, W. Lin, M. Hong, K.A. Loparo, Voltage regulation of DC-microgrid with PV and battery: A passivity method, IFAC- Pap. 52 (16) (2019) 753–758, http://dx.doi.org/10.1016/j.ifacol.2019.12.053.

[27] M.M. Eskander, C.A. Silva, Techno-economic and environmental comparative analysis for DC microgrids in households: Portuguese and french household case study, Appl. Energy 349 (2023) 121495, http://dx.doi.org/10.1016/j.apenergy.2023.121495.

[28] M. Fantauzzi, D. Lauria, F. Mottola, A. Scalfati, Sizing energy storage systems in DC networks: A general methodology based upon power losses minimization, Appl. Energy 187 (2017) 862–872, http://dx.doi.org/10.1016/j.apenergy.2016.11.044.

[29] A. Molina, O.D. Montoya, W. Gil-González, Exact minimization of the energy losses and the CO2 emissions in isolated DC distribution networks using PV sources, DYNA 88 (217) (2021) 178–184, http://dx.doi.org/10.15446/dyna.v88n217.93099.

[30] L.F. Grisales-Noreña, J.A. Ocampo-Toro, O.D. Montoya-Giraldo, J. Montano, J. Hernandéz, Optimal operation of battery storage systems in standalone and grid-connected DC microgrids using parallel metaheuristic optimization algorithms, J. Energy Storage 65 (2023) 107240, http://dx.doi.org/10.1016/j.est.2023.107240.

[31] Y. Yusoff, M.S. Ngadiman, A.M. Zain, Overview of NSGA-II for optimizing machining process parameters, Procedia Eng. 15 (2011) 3978–3983, http://dx.doi.org/10.1016/j.proeng.2011.08.745.

[32] Y. Hu, Z. Bie, T. Ding, Y. Lin, An NSGA-II based multi-objective optimization for combined gas and electricity network expansion planning, Appl. Energy 167 (2016) 280–293, http://dx.doi.org/10.1016/j.apenergy.2015.10.148.

[33] H. Liu, X. Xing, W. Shang, T. Li, NSGA-II optimized multiobjective predictive energy management for fuel cell/battery/supercapacitor hybrid construction vehicles, Int. J. Electrochem. Sci. 16 (4) (2021) 21046, http://dx.doi.org/10.20964/2021.04.24.

[34] X. Wen, Q. Song, Y. Qian, D. Qiao, H. Wang, Y. Zhang, H. Li, Effective improved NSGA-II algorithm for multi-objective integrated process planning and scheduling, Mathematics 11 (16) (2023) 3523, http://dx.doi.org/10.3390/math11163523.

[35] P. Zhang, Y. Qian, Q. Qian, Multi-objective optimization for materials design with improved NSGA-II, Mater. Today Commun. 28 (2021) 102709, http://dx.doi.org/10.1016/j.mtcomm.2021.102709.

[36] S. Verma, M. Pant, V. Snasel, A comprehensive review on NSGA-II for multi-objective combinatorial optimization problems, IEEE Access 9 (2021) 57757–57791, http://dx.doi.org/10.1109/access.2021.3070634.

[37] M.T. Emmerich, A.H. Deutz, A tutorial on multiobjective optimization: fundamentals and evolutionary methods, Nat. Comput. 17 (2018) 585–609.

[38] L.F. Grisales-Noreña, O.D. Montoya, C.A. Ramos-Paja, An energy management system for optimal operation of BSS in DC distributed generation environments based on a parallel PSO algorithm, J. Energy Storage 29 (2020) 101488.

[39] A. Soroudi, Power System Optimization Modeling in GAMS, vol. 78, Springer, 2017.

[40] M.S. Javadi, M. Lotfi, M. Gough, A.E. Nezhad, S.F. Santos, J.P. Catalao, Optimal spinning reserve allocation in presence of electrical storage and renewable energy sources, in: 2019 IEEE International Conference on Environment and Electrical Engineering and 2019 IEEE Industrial and Commercial Power Systems Europe (EEEIC/I&CPS Europe), IEEE, 2019, pp. 1–6.

[41] K. Deb, A. Pratap, S. Agarwal, T. Meyarivan, A fast and elitist multiobjective genetic algorithm: NSGA-II, IEEE Trans. Evol. Comput. 6 (2002) 182–197, http://dx.doi.org/10.1109/4235.996017.

[42] B. Cortés-Caicedo, F. Molina-Martin, L.F. Grisales-Noreña, O.D. Montoya, J.C. Hernández, Optimal design of PV systems in electrical distribution networks by minimizing the annual equivalent operative costs through the discrete-continuous vortex search algorithm, Sensors 22 (3) (2022) 851.

[43] J.H. Holland, Adaptation in Natural and Artificial Systems, MIT Press, 1975.

[44] R.I. Bolaños, R.A.H. Isaza, R.A.G. Rendón, Reubicación óptima de transformadores de distribución usando el algoritmo multiobjetivo de optimización NSGA II, Ing. Cienc. 10 (2014) 77–98.

[45] M.S. Javadi, C.S. Gouveia, L.M. Carvalho, R. Silva, Optimal power flow solution for distribution networks using quadratically constrained programming and McCormick relaxation technique, in: 2021 IEEE International Conference on Environment and Electrical Engineering and 2021 IEEE Industrial and Commercial Power Systems Europe (EEEIC/I&CPS Europe), IEEE, 2021, pp. 1–6.

[46] M.S. Javadi, M. Gough, S.A. Mansouri, A. Ahmarinejad, E. Nematbakhsh, S.F. Santos, J.P. Catalao, A two-stage joint operation and planning model for sizing and siting of electrical energy storage devices considering demand response programs, Int. J. Electr. Power Energy Syst. 138 (2022) 107912.

[47] L.F. Grisales-Noreña, J.C. Morales-Duran, S. Velez-Garcia, O.D. Montoya, W. Gil-González, Power flow methods used in AC distribution networks: An analysis of convergence and processing times in radial and meshed grid configurations, Results Eng. 17 (2023) 100915.

[48] L.F. Grisales-Noreña, O.D. Montoya, B. Cortés-Caicedo, F. Zishan, J. Rosero-García, Optimal power dispatch of pv generators in ac distribution networks by considering solar, environmental, and power demand conditions from Colombia, Mathematics 11 (2) (2023) 484.

[49] R. Cheraghi, M.H. Jahangir, Multi-objective optimization of a hybrid renewable energy system supplying a residential building using NSGA-II and MOPSO algorithms, Energy Convers. Manage. 294 (2023) 117515.

[50] R. Sellami, S. Farooq, N. Rafik, An improved MOPSO algorithm for optimal sizing & placement of distributed generation: A case study of the Tunisian offshore distribution network (ASHTART), Energy Rep. 8 (2022) 6960–6975.

[51] M.Z. Malik, M. Kumar, A.M. Soomro, M.H. Baloch, M. Farhan, M. Gul, G.S. Kaloi, Strategic planning of renewable distributed generation in radial distribution system using advanced MOPSO method, Energy Rep. 6 (2020) 2872–2886.

[52] X. Lv, Y. Lv, Y. Zhu, Multi-variable study and MOPSO-based multi-objective optimization of a novel cogeneration plant using biomass fuel and geothermal energy: A complementary hybrid design, Energy 270 (2023) 126921.

[53] Z. Belboul, B. Toual, A. Kouzou, A. Bensalem, Optimization of hybrid PV/Wind/Battery/DG microgrid using MOALO: A case study in djelfa, Algeria, in: 2022 19th International Multi-Conference on Systems, Signals & Devices, SSD, IEEE, 2022, pp. 1615–1621.

[54] C. Dhifaoui, I. Marouani, H.H. Abdallah, MOALO algorithm applied to dynamic economic environmental dispatch including renewable energy, IJCSNS 20 (6) (2020) 36–47.

[55] A. Kazikova, M. Pluhacek, R. Senkerik, Why tuning the control parameters of metaheuristic algorithms is so important for fair comparison? in: Mendel, vol. 26, 2020, pp. 9–16, 2.

[56] A. Esmaeel Nezhad, A. Ahmadi, M.S. Javadi, M. Janghorbani, Multi-objective decision-making framework for an electricity retailer in energy markets using lexicographic optimization and augmented epsilon-constraint, Int. Trans. Electr. Energy Syst. 25 (12) (2015) 3660–3680.

[57] Y. Jiang, L. Kang, Y. Liu, Multi-objective design optimization of a multi-type battery energy storage in photovoltaic systems, J. Energy Storage 39 (2021) 102604.